MATHEMATICAL FORMULAE

1 Vector Algebra

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} = (a_1, a_2, a_3)$$

Scalar(Dot) Product $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$

Vector (Cross) Product

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}.$$

Triple vector product $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$

Triple scalar product

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_1b_3c_2 - a_2b_1c_3 - a_3b_2c_1$$

2 Series

$$(1+x)^n = 1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\ldots+\frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}x^r+\ldots \ (n \text{ arbitrary}, |x|<1)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \ldots + \frac{x^r}{r!} + \ldots \qquad \text{for all } x$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots + (-1)^{r+1}\frac{x^r}{r} + \ldots \qquad \text{for } -1 < x \leq 1$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots + \frac{(-1)^r x^{2r}}{(2r)!} + \ldots \qquad \text{for all } x$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots + \frac{(-1)^r x^{2r+1}}{(2r+1)!} + \ldots \qquad \text{for all } x$$

3 Trigonometric Identities and Hyperbolic Functions

$$\sin(A+B) = \sin A\cos B + \sin B\cos A$$

$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\cos(iz) = \cosh z\,, \quad \cosh(iz) = \cos z\,, \quad \sin(iz) = i\,\sinh(z)\,, \quad \sinh(iz) = i\,\sin z$$

4 Differential Calculus

1. Leibnitz' Formula: $D^n(fg) =$

$$D^n f\, g + \binom{n}{1} D^{n-1} f\, Dg + \binom{n}{2} D^{n-2} f\, D^2 g + \cdots + \binom{n}{r} D^{n-r} f\, D^r g + \cdots + f D^n g$$

where $\begin{pmatrix} n \\ r \end{pmatrix} = \dfrac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$.

2. Taylor expansion of $f(x)$ about the point $x = a$

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$$

where

$$R_n(x) = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(c), \qquad a < c < x$$

Putting $x = a + h$ gives the form

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \ldots$$

3. Taylor expansion of $f(x, y)$ about the point (a, b):

$$f(x,y) = f(a,b) + \frac{1}{1!}\left[(x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right] f(a,b) + \frac{1}{2!}\left[(x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right]^2 f(a,b) + \cdots$$

4. Chain rule for partial derivatives: $f = f(x, y)$

(i) If $x = x(t)$ and $y = y(t)$ then $f = F(t)$ and

$$\frac{dF}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

(ii) If x and y are expressed in terms of new variables u and v by $x(u, v)$ and $y(u, v)$ then $f = F(u, v)$ and

$$\frac{\partial F}{\partial u} = \frac{\partial f}{\partial x}\,\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\,\frac{\partial y}{\partial u}, \quad \text{and} \quad \frac{\partial F}{\partial v} = \frac{\partial f}{\partial x}\,\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\,\frac{\partial y}{\partial v}$$

(iii) If $y = y(x)$ then $f = F(x)$ and

$$\frac{dF}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx}$$

5. Stationary points of $f(x, y)$ occur where $f_x = 0$ and $f_y = 0$. Let (a, b) be a stationary point, and consider $D = f_{xx}(a,b) f_{yy}(a,b) - [f_{xy}(a,b)]^2$. Then

   (a) If $D > 0$ and $f_{xx}(a, b) < 0$, then (a, b) is a maximum;
   (b) If $D > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a minimum;
   (c) If $D < 0$ then (a, b) is a saddle point;
   (d) if $D = 0$, more information is required.

6. Differential Equations

(i) The first order linear equation $\dfrac{dy}{dx} + p(x)y = r(x)$ has an integrating factor $I(x) = \exp\left\{\displaystyle\int p(x)\, dx\right\}$ which allows one to write the DE in the form $\dfrac{d}{dx}(I(x)y) = I(x)r(x)$.

(ii) The first-order equation $P(x, y)\dfrac{dy}{dx} + Q(x, y) = 0$ is exact if $\dfrac{\partial P}{\partial x} = \dfrac{\partial Q}{\partial y}$.

5 Integral Calculus

1. An important substitution: $\tan(\theta/2) = t$; then

$$\sin\theta = \frac{2t}{1+t^2}\,, \quad \cos\theta = \frac{1-t^2}{1+t^2}\,, \quad d\theta = \frac{2\ dt}{1+t^2}\,.$$

2.
$$\int \frac{dx}{(a^2-x^2)^{1/2}} \;=\; \sin^{-1}\left(\frac{x}{a}\right)\,, \quad |x| < a\,.$$

3.
$$\int \frac{dx}{(a^2+x^2)^{1/2}} \;=\; \sinh^{-1}\left(\frac{x}{a}\right) \;=\; \ln\left[\frac{x}{a} + \left(\frac{x^2}{a^2}+1\right)^{1/2}\right]\,.$$

4.
$$\int \frac{dx}{(x^2-a^2)^{1/2}} \;=\; \cosh^{-1}\left(\frac{x}{a}\right) \;=\; \ln\left[\frac{x}{a} + \left(\frac{x^2}{a^2}-1\right)^{1/2}\right]\,.$$

5.
$$\int \frac{dx}{a^2+x^2} \;=\; \left(\frac{1}{a}\right)\tan^{-1}\left(\frac{x}{a}\right)\,.$$

6 Fourier Series

The following formulae assume that $f(x)$ satisfies the Dirichlet conditions and is periodic with period T, i.e. $f(x+T) = f(x)$. The general Fourier Series for $f(x)$ is

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\left[a_n\cos\left(\frac{2n\pi x}{T}\right) + b_n\sin\left(\frac{2n\pi x}{T}\right)\right]$$

where

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(x)\ \cos\left(\frac{2n\pi x}{T}\right)\,dx\,,\ b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(x)\ \sin\left(\frac{2n\pi x}{T}\right)\,dx\,,\ n = 0,1,2,3,\ldots$$

The series converges to $f(x)$ at points of continuity and to the mean value $\frac{1}{2}(f(x_+)+f(x_-))$ at points where $f(x)$ is discontinuous.

The complex form of the Fourier series is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nx/T}\,, \text{ where } c_n = \frac{1}{T}\int_{-T/2}^{T/2} f(x)e^{-i2n\pi x/T}\,dx \text{ for every integer } n.$$

Half-range series: If $f(t)$ is an even (resp. odd) function, all sine (resp. cosine) terms vanish, and we have a half-range cosine (resp. sine) series. Let $L = T/2$. Then the coefficients are, respectively:

$$f(t) \text{ even: } b_n = 0\,,\ a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx\,. \qquad f(t) \text{ odd: } a_n = 0\,,\ b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx\,.$$

Parseval's Theorem: If the complex Fourier series of the T-periodic function $f(x)$ has coefficients c_n, and the real Fourier series of $f(x)$ has coefficients a_n and b_n, then

$$\frac{1}{T}\int_{-T/2}^{T/2} f^2(x)\ dx \;=\; \sum_{n=-\infty}^{\infty}|c_n|^2\,. \qquad \text{and} \qquad \frac{2}{T}\int_{-T/2}^{T/2} f^2(x)\ dx = \frac{1}{2}a_0^2 + \sum_{n=1}^{\infty} a_n^2 + b_n^2\,.$$

7 Fourier Transforms

The following formulae assume that $g(t)$ and $f(t)$ are the sum of one or more components that obey the Dirichlet conditions. The Fourier transform pair is

$$G(\omega) = \int_{-\infty}^{\infty} g(t) e^{-i\omega t}\, dt\,, \quad \Longleftrightarrow \quad g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{i\omega t}\, d\omega\,,$$

where $G(\omega) = \mathcal{F}\left[g(t)\right]$ is the *Fourier transform* of $g(t)$, while $g(t) = \mathcal{F}^{-1}\left[G(\omega)\right]$ is the *inverse Fourier transform* of $G(\omega)$.

The Dirac delta function $\delta(t)$ satisfies

$$\int_{-\infty}^{\infty} \delta(t) dt = 1, \qquad \int_{-\infty}^{\infty} f(t)\delta(t-a) dt = f(a)$$

For a periodic function $f(t)$ with angular frequency ω_0, and complex Fourier series as given,

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{in\omega_0 t} \qquad \Longrightarrow \qquad \mathcal{F}\left[f(t)\right] = F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_0)\,.$$

Time- and Frequency-shift: If $\mathcal{F}\left[f(t)\right] = F(\omega)$ then

$$\mathcal{F}[f(t-t_0)] = e^{-i\omega t_0} F(\omega) \qquad \text{and} \qquad \mathcal{F}\left[e^{i\omega_0 t} f(t)\right] = F(\omega - \omega_0)\,. \tag{1}$$

Symmetry/Duality property: $\mathcal{F}\left[g(t)\right] = F(\omega) \Longleftrightarrow \mathcal{F}\left[G(t)\right] = 2\pi g(-\omega)$